

# Late Breaking Results: Less Sense Makes More Sense: In-Sensor Compressive Learning for Efficient Machine Vision

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**Abstract**—Integrating deep learning and image sensors has significantly transformed machine vision applications. Yet, conventional high-resolution image acquisition schemes enabled by imagers are energy-inefficient for deep learning, as they involve excessive data quantization and transmission overhead. To address this challenge, we propose a lightweight in-sensor compressive learning framework that integrates a compressive learning-based encoder within image sensors for task-specific feature extraction. Our framework encodes raw images into adaptive low-dimensional representations using only a 1-bit encoder by joint optimization with downstream machine vision tasks. It achieves  $10\times$  data compression, a minimum of 1.6% accuracy loss in the task, and  $3.93\times$  energy savings at the sensor-end, outperforming prior arts.

## I. INTRODUCTION AND BACKGROUND

Integrating deep learning with image sensors has revolutionized machine vision with unprecedented accuracy by shifting decision-making from human-oriented analysis of high-quality images to learning-enabled feature extraction directly from sensory data. Yet, conventional vision sensors still capture high-fidelity images not efficient features for ubiquitous machine vision tasks, leading to high energy use in sensor front end due to massive data quantization and transmission. To address this challenge, various energy-efficient methods have been explored at the edge, including image compression and in-sensor processing. Traditional compression methods (e.g., JPEG and compressive sensing), reduce transmission costs but remain task-agnostic and inefficient for real-time learning applications [1], [2]. In-sensor processing integrates sensing with learning to mitigate the challenge but introduces resource overheads and inflexibility to sensors [3], [4]. This “late breaking results” presents a lightweight in-sensor compressive learning framework for efficient machine vision tasks. It integrates a learnable encoder within image sensors to efficiently extract task-relevant features for downstream tasks through joint optimization with a deep neural network (DNN) backbone. Specifically, the encoder can be implemented in image sensors with their inherent computing resources without introducing any overhead.

Our study shows that a 1-bit encoder can be developed to enable  $10\times$  data compression with a minimum of 1.6% accuracy loss in the given task and achieve  $3.93\times$  energy savings at the sensor-end, outperforming state-of-the-art compressive methods. Our work provides an intriguing insight into “Less Sense Makes More Sense” for efficient edge machine vision tasks.

## II. IN-SENSOR COMPRESSIVE LEARNING

The proposed in-sensor compressive learning framework comprises two key components: (1) A jointly trained encoder-decoder optimized with backbone vision models for downstream tasks; (2) A digital implementation of the encoder without requiring extra sensor hardware overhead. As shown in Fig. 1(a), the encoder applies a block-based compressive learning approach with a learnable sampling matrix to extract task-relevant features. It encodes images into low-dimensional representations, quantized with an adjustable bit depth  $Q$ . Unlike conventional generative AI methods (e.g., auto-encoders) that requires complex decoders for high image quality, a simple linear projection suffices to reconstruct essential features at the original

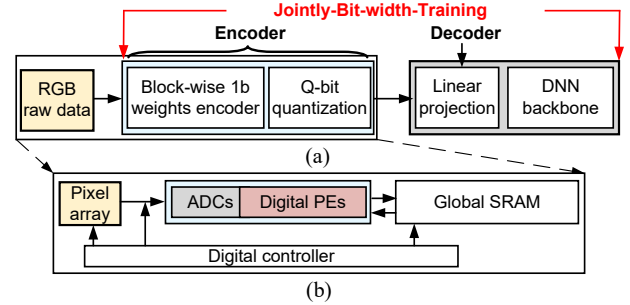


Fig. 1. (a) Whole framework processing pipeline. (b) Digital in-sensor implementation.

image size for seamless integration with backbone models. At the hardware level, we propose to embed the encoder into sensors to minimize data transmission and reduce energy consumption. The system (Fig. 1(b)) includes a pixel array, column-parallel readout circuits, analog-to-digital converters (ADCs), processing elements (PEs), memory, and a controller. The PEs process the incoming data by utilizing the encoder weight matrix from global SRAM to produce quantized output features, which are then stored back in global SRAM for off-chip transmission. A digital controller coordinates data flow from exposure to final readout. Note that PEs and memory are repurposed by harnessing the inherent computing resources within the conventional image sensor processing pipeline.

### A. Design of the Learnable Compressive Encoder

The encoder employs a block-based compression strategy, dividing images into  $k \times k$  blocks for efficient processing (Fig. 2(a)). This approach allows compression to be performed on a block-by-block basis using a single encoder, significantly reducing memory requirements compared to traditional compressive sensing methods [5] that utilize a large sampling matrix for the entire image. With a compressive ratio  $r$ , each block  $x_i$  is compressed with a learnable matrix  $\Phi$ , yielding output  $y_i = \Phi x_i$ , where  $\Phi$  is an  $rk^2 \times k^2$  matrix. This is similar to depth-wise convolution but with a shared kernel across channels. The decoder reconstructs task-relevant features using a simple linear projection, leveraging the transposed sampling matrix for efficient feature recovery. To enhance hardware efficiency, we binarize the sampling matrix. Specifically, during the training, we binarize the sampling matrix  $\Phi$ ,  $\Phi_b = B(\Phi)$  as

$$B(\alpha) = \begin{cases} 1, & \text{if } \alpha \geq 0; \\ -1, & \text{if } \alpha < 0. \end{cases} \quad (1)$$

Here, function  $B$  is an element-wise operation, and the gradient is denoted by

$$B'(\alpha) = \begin{cases} 1, & \text{if } -1 \leq \alpha \leq 1; \\ 0, & \text{otherwise.} \end{cases} \quad (2)$$

Thus,  $B(\alpha)$  is used in the forward pass, while  $B'(\alpha)$  is employed in back-propagation. A universal matrix encodes all three image channels in-sensor for consistency, while channel-specific scaling parameters are applied off-chip for task-specific optimization. The

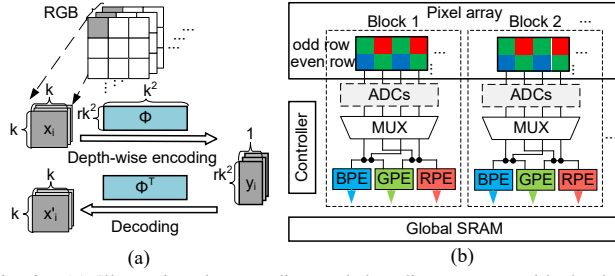


Fig. 2. (a) Illustrating the encoding and decoding process with depth-wise encoder structure. (b) Data flow of row exposure for a  $2 \times 2$  encoder under Bayer pattern pixel array.

encoder-decoder network is jointly optimized with the backbone model to ensure end-to-end task accuracy.

### B. In-Sensor Digital Implementation of the Encoder

Fig. 2(b) illustrates the encoder's data flow. The encoder processes images block-by-block, utilizing a shared sensing matrix across all three color channels. Thus, the operations performed by each PE are identical for each channel. In Bayer-pattern image sensors, each row captures either RG (Red-Green) or BG (Blue-Green) pixels, which are assigned to their respective PEs. As each row is exposed, pixel values are directed to their PEs for accumulation. Once all rows within a block are processed, output feature maps are generated. The digital PEs can be implemented by reusing computing resources in the conventional processing pipeline of image sensors, which performs addition/subtraction with binary weights (1 or  $-1$ ) to accumulating partial sums across exposure cycles. After processing all rows in a block, the final result is quantized to  $Q$  bit. Therefore, the total compression ratio (CR) is defined as  $CR = Q_{full}/(r \cdot Q)$ .  $Q_{full}$  represents the bit depth of original pixels (i.e., 8 bit). Note that many modern image sensors (e.g., Sony IMX500/IMX501, ON Semiconductor AR0430, OmniVision OV580, and others) are equipped with a reconfigurable digital signal processing pipeline, enabling them to be repurposed as PEs without incurring additional area overhead.

## III. EXPERIMENTAL METHODOLOGY AND RESULTS

To comprehensively assess the energy efficiency of our proposed framework, we develop a modeling framework by leveraging the experimental data reported by previous work under CMOS 65 nm technology. Specifically, we decompose the energy consumption at the block level into three primary components. **Pixel Readout:** Exposure and readout consume 12.1 pJ per pixel based on prior studies [6]. **PE Computation & ADC:** Based on the current CMOS image sensor [7], the energy required for a 10-bit ADC is approximately 106.08 pJ, halving per bit reduction. For PEs, the number of 1-b MAC operations is dictated by the sampling matrix size  $rk^2 \times k^2$ . We estimate this energy using a full adder (FA) model. **Off-chip transmission:** Transmitting one byte via MIPI CSI-2 costs approximately 100 pJ [8]. Local SRAM access (10 pJ per 64-bit at 45 nm [9]) is scaled to 65 nm. By exploring the design space, we find that using 4-bit raw pixels and 8-bit features with a 0.1 compressive ratio achieves the optimal balance between accuracy and energy efficiency.

We compare our framework with other compression methods such as low-resolution quantization (LR), and spatial down-sampling (SD). Among various vision tasks, image classification on ImageNet (384  $\times$  384 resolution) is chosen as the benchmark task. A block size of 16  $\times$  16 is used to balance the compressive ratio and the compute unit requirements. To suit edge vision applications, we use a pre-trained TinyViT as the backbone DNN. Quantization-aware

TABLE I  
ACCURACY AND ENERGY EFFICIENCY VS. CNV-CIS.

| Method                 | Ours          | LR            | SD            |
|------------------------|---------------|---------------|---------------|
| Compression ratio (CR) | 10            | 4             | 4             |
| Accuracy loss          | 1.6%          | 26.5%         | 22%           |
| Energy efficiency      | 3.93 $\times$ | 2.14 $\times$ | 2.23 $\times$ |

training is applied to both raw pixels and features. A conventional CIS sensor (CNV-CIS) with 8-bit pixels is used as a baseline. For the baseline, the raw images generated by such a sensor is directly used by the backbone. This baseline sensor achieves an energy consumption of 151.5 pJ/pixel with an accuracy of 85%. Table I presents the results, where all values are normalized to the baseline. The comparison demonstrates that our approach achieves superior performance, offering substantial energy savings while maintaining the highest compression ratio with minimal accuracy degradation. This improvement is primarily attributed to significant reductions in ADCs lower-bit quantization, simplified 1-bit computations, and communication energy. Moreover, joint training with the backbone enables the model to learn task-specific features, facilitating efficient and adaptive learning-enabled image sensors for edge applications by optimizing feature extraction.

## IV. CONCLUSION AND FUTURE WORKS

This work proposes a compressive learning paradigm for edge vision that embeds a learnable encoder within sensors. Jointly trained with a backbone DNN, our approach achieves a 10 $\times$  compression ratio and minimal relative accuracy loss (1.6%) with 3.93 $\times$  energy savings. **For future work**, we intend to develop an analog implementation, expand to diverse vision tasks with various models, and conduct real hardware evaluations using commercial image sensors (e.g., OmniVision OV580) to validate performance and efficiency.

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